

SEBA-XAI: Explainable Policy-Aware Audit for Secure Inter-Agency Police Data Access Governance

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Abstract—In criminal-justice workflows, job role alone is not sufficient to decide access to a sensitive record. For example, a sub-inspector and a constable may request the same case file and should not receive the same answer; the same officer may be permitted for one purpose and denied for another; and some records must remain closed to one department while another may read them. This paper proposes SEBA-XAI, an explainable access-governance overlay for record sharing between agencies. For every request the system evaluates attributes of the requester, the record, the requested action, the stated purpose and the operating context, and returns allow, deny or escalate. It also stores the grounds for the decision: the attributes that were decisive and the policy version applied. Both are recorded in a tamper-evident audit trail on a permissioned blockchain that multiple agencies can verify, while the record remains with the agency holding it. The paper also studies an attack that such audit trails cannot detect. If an attacker controls the component that signs audit entries, a decision can be modified and signed again: the entry passes verification although the stored decision was never authorised. Since real access logs are not public, evaluation uses a reproducible synthetic workload over five seeds. The results show that all integrity-based defences considered detect ordinary tampering but obtain 0.00 detection for this attack, whereas two methods that check the decisions instead of the signatures obtain 1.00 detection under stronger visibility assumptions. Integrity and correctness are therefore separate properties, and tamper detection alone is not sufficient to establish trust in an audit trail.

Index Terms—Access control, access governance, attribute-based access control, blockchain audit, explainable AI, inter-agency data sharing, tamper evidence, auditability.

I. Introduction

CONSIDER an armed robbery registered at a town police station. Three men rob a jewellery shop at closing time, an employee is injured, and one robber leaves blood on a display counter. Within two weeks the case file contains the first information report, the complainant and witness statements, a medico-legal certificate, the investigating officer’s case diary, seizure memos, CCTV footage, forensic exhibits, and the call records of a stolen phone. During investigation, one of the accused is found to be seventeen years old, so his identity and records fall under statutory juvenile protection. From this point, requests for the file arrive from several directions, and each one needs a different answer.

The sub-inspector assigned as investigating officer needs the full file and should receive it. A constable from the same station, with no connection to the case, asks for the case diary and should be refused, although his station and his role are the same as his colleague’s. A sub-inspector from a neighbouring district, investigating a similar gang robbery, requests the CCTV stills to compare the method of operation; his purpose is legitimate, but the request crosses a district boundary, so it should be neither silently granted nor silently refused. It should go to a supervisory officer, and the approval itself should be recorded. A forensic scientist analysing the blood sample needs the exhibit-chain records and should get them, but when the same scientist requests the victim’s name and address a few minutes later, the answer must be no, because nothing in a forensic examination requires it. The superintendent of police, despite outranking everyone above, must be refused the juvenile’s identity, because a statutory protection class is stronger than rank. Finally, six months later, an inquiry authority who was present for none of these events must be able to verify what was released, to whom, on what grounds, and who approved it.

This scenario is the subject of the paper, and it shows three things that any access mechanism for such workflows must provide. First, the decision cannot be derived from the job role alone: the constable and the investigating officer hold comparable roles and receive opposite answers, and the forensic scientist receives opposite answers to two of his own requests made minutes apart. The decision depends on attributes of the requester, the record, the requested action, the stated purpose, and the operating context. Second, allow and deny are not enough: some requests are legitimate but cross a boundary that requires a human authority, so escalate must be a first-class outcome and the resulting approval must become part of the evidence. Third, decision evidence must outlive the decision: whatever the inquiry authority needs in month six must have been recorded in month one, because it cannot be reconstructed afterwards. The risk of weak governance here is not hypothetical. In 2023, a data breach at the Police Service of Northern Ireland exposed the names, ranks, and locations of serving officers, and the UK Information Commissioner’s Office later imposed a

fine over the incident [1], [2].

The infrastructure to move such records between agencies already exists. In India, the Crime and Criminal Tracking Network and Systems (CCTNS) connects police stations nationwide, and the Inter-Operable Criminal Justice System (ICJS) links the police pillar with courts, prisons, forensic laboratories, and prosecution [3], [4]. This paper does not propose a replacement for these systems. It addresses what they leave open: once a record is technically reachable across an agency boundary, the decision to release it must be made from context, recorded with its grounds, and protected against later manipulation.

Existing work covers parts of this requirement, and each part stops short at a known point. Role-based access control assigns permissions to job titles [5], but the scenario above defeats it in its first two requests. Attribute-based access control decides from subject, object, action, and environment attributes [6], and recent work extends it toward accountability [7] and privacy preservation [8]; policy engines of this kind, however, are built to produce decisions, not to preserve reviewable evidence of how each decision was reached. Blockchain-based audit preserves evidence in a tamper-evident, multi-party form [9], [10] and has been applied to digital evidence protection [11], [12], but it certifies only that a stored entry has not been altered after it was written. Explainable AI produces reasons for decisions, and the argument that high-stakes systems should prefer reviewable decisions is well established [13], [14]; in practice, however, explanations are rendered for the requester and discarded, rather than stored as evidence bound to the decision they justify.

There is also a failure mode that none of the audit mechanisms above can detect, and the scenario makes it concrete. Suppose the denial issued to the forensic scientist for the victim's identity is later rewritten into an approval by an attacker who controls the component that signs audit entries. The rewritten entry is signed with the genuine key, so its signature verifies, the chain remains intact, and every agency's copy of the ledger agrees. The inquiry authority in month six runs an integrity check and finds nothing wrong, yet the ledger now preserves a decision that policy never made. Integrity and correctness are separate properties: a signature answers whether the log was changed after writing, not whether what was written was true.

This paper proposes SEBA-XAI, a Secure, Explainable, Blockchain-Audited access-governance overlay that addresses the scenario end to end. For every request, a policy service evaluates the attributes above and returns allow, deny, or escalate. At decision time, the system records the grounds of the decision: the attributes that were decisive and the policy version applied. The decision and its grounds are committed to a permissioned blockchain audit trail shared by the participating agencies, so that no single agency can silently rewrite history, while raw records never leave the agency holding them. Because the re-signing attack defeats even this audit trail from the inside, the paper additionally evaluates detection methods

that examine the recorded decisions rather than their signatures.

A. Practical Challenges

The paper is organised around four practical challenges taken directly from the scenario. Each challenge is paired with one contribution in the same order.

- Contextual access governance. Access cannot be decided from static roles: the constable and the investigating officer share a role but not an assignment, and the forensic scientist's two requests differ only in the record and purpose. The decision must consider the requester, record, action, purpose, approval state, and operating context.
- Tamper-evident audit across agencies. The inquiry authority can trust the decision evidence only if no agency could have silently changed it, and raw sensitive records must not be copied to a shared ledger to achieve this.
- Explainable access decisions. Every allow, deny, or escalate outcome in the scenario must carry its grounds, because these decisions affect investigation timelines, privacy protection, inter-agency accountability, and later legal or administrative review.
- Valid-looking corrupted logs. An attacker who controls a signing component can rewrite a decision and re-sign it, so the audit trail remains internally valid while recording a decision that was never made.

B. Research Contributions

To address these challenges, the paper makes four contributions, ordered to match the challenges above.

- Contribution 1: A layered access-control service that combines role-based, attribute-based, and policy-based rules and returns allow, deny, or escalate for inter-agency record requests, so that cases such as the constable's request and the forensic scientist's second request are decided from context rather than role.
- Contribution 2: A permissioned blockchain audit layer in which each decision event records the request identifier, decision hash, explanation hash, policy version, approval reference where one exists, and record commitment, while raw records remain off-chain with the holding agency.
- Contribution 3: An explainable decision service that stores the decisive attributes, reason code, policy version, and counterfactual information as a structured artifact hash-linked to the audit event, so that a reviewer can verify the grounds of a decision without opening the underlying records.
- Contribution 4: A reproducible adversarial benchmark covering ordinary tampering attacks and the validly re-signed compromised-signer attack. Across five seeds of a synthetic workload, every integrity-based defence tested detects ordinary tampering but obtains 0.00 detection for the re-signed attack, while two methods that examine the decisions themselves,

a statistical drift detector over the decision log and a trusted attribute oracle, obtain 1.00 detection under stronger visibility assumptions.

The scope of the claims is deliberately limited. Real police access logs and records are not publicly available, so the evaluation uses a reproducible synthetic workload, and the blockchain layer is a simulated permissioned ledger rather than a live deployment. The paper does not claim deployment inside any operational system, legal compliance, validation on real police records, or production security. It also deliberately avoids crime prediction: aggregate public crime statistics are not individual-level access-control data [15], predictive policing can create feedback loops when observed crime data is shaped by earlier policing decisions [16], and complex criminal-justice prediction systems do not reliably outperform simple baselines [17]. The subject of this paper is the governance of access, not the prediction of crime.

The remainder of the paper is organised as follows. Section ?? reviews related work. Section ?? presents the SEBA-XAI architecture, threat model, and benchmark design. Section ?? reports the results, including the compromised-signer asymmetry. Section ?? states the limitations, and Section ?? concludes.

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